

Programming Test Results (Without Test Cases)

Result Summary

Field	Value
Test ID	41406
Student ID	29227
Programs (without test cases)	2
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Programs Without Test Cases

#	Program Name	Status	Submitted At
1	HashSetToArray	Submitted	31/01/2026, 10:27:33
2	ExtractKeysfromHashMap	Submitted	31/01/2026, 10:26:01

Program Details (Without Test Cases)

Program 1: HashSetToArray

Languages:	Java
Status:	Submitted
Submitted At:	31/01/2026, 10:27:33
Description:	The objective of this program is to accept multiple elements from the user, store them in a HashSet to ensure uniqueness, convert the HashSet into an array, and display all the array elements.
Constraints:	-
Sample Input:	Enter number of elements: 3 Java Programming Data Structures Operating Systems
Sample Output:	Array Elements: Java Programming Data Structures Operating Systems
Explanation:	Import necessary packages: Use java.util.HashSet and java.util.Scanner. Create a HashSet to store elements (e.g., String elements). Use Scanner to take input from the user for multiple elements. Add elements to the HashSet to ensure duplicates are removed automatically. Convert the HashSet into an array Iterate through the array using a loop. Display each element of the array.

Solution Code

```

import java.util.HashSet.*;
void main()
{
    HashSet<String> h = new HashSet<>(4);
    for(int i=0;i<=2;i++)
    {
        String s1 = IO.readLine();
        h.add(s1);
    }

    IO.println("Array Elemets:");
    ArrayList<String> ar = new ArrayList<>(h);
    for(String s : ar)
    {

        IO.println(s);
    }

}

```

Program 2: ExtractKeysfromHashMap

Languages:	Java
Status:	Submitted
Submitted At:	31/01/2026, 10:26:01
Description:	The objective of this program is to accept product details (Product ID and Product Name) from the user, store them in a HashMap, and then extract and display all the product IDs stored as keys in the map.
Constraints:	-
Sample Input:	Enter number of products: 3 Enter Product ID: 101 Enter Product Name: Pen Enter Product ID: 102 Enter Product Name: Book Enter Product ID: 103 Enter Product Name: Pencil
Sample Output:	List of Product Keys: 101 102 103
Explanation:	Create a Product class with productId and productName. Create a HashMap<Integer, Product> to store product details. Use Scanner to accept user input. Ask the user for the number of products. Loop through the input and: Read product ID and product name. Store them in the HashMap. Extract keys Display all extracted product IDs.

Solution Code

```

import java.util.Map.*;

void main()
{

```

```
Map<String,String> m = new HashMap<>();
int n = Integer.parseInt(IO.readLine("Enter size"));
for(int i=0;i<n;i++)
{
    //Integer id = Integer.parseInt("Enter Product ID:\n");
    String id = IO.readLine("");
    String name = IO.readLine("Enter Product Name");
    m.put(id,name);

}
IO.println("\nList of Product Keys:");
m.forEach((k,v)->IO.println(k));
}
```